

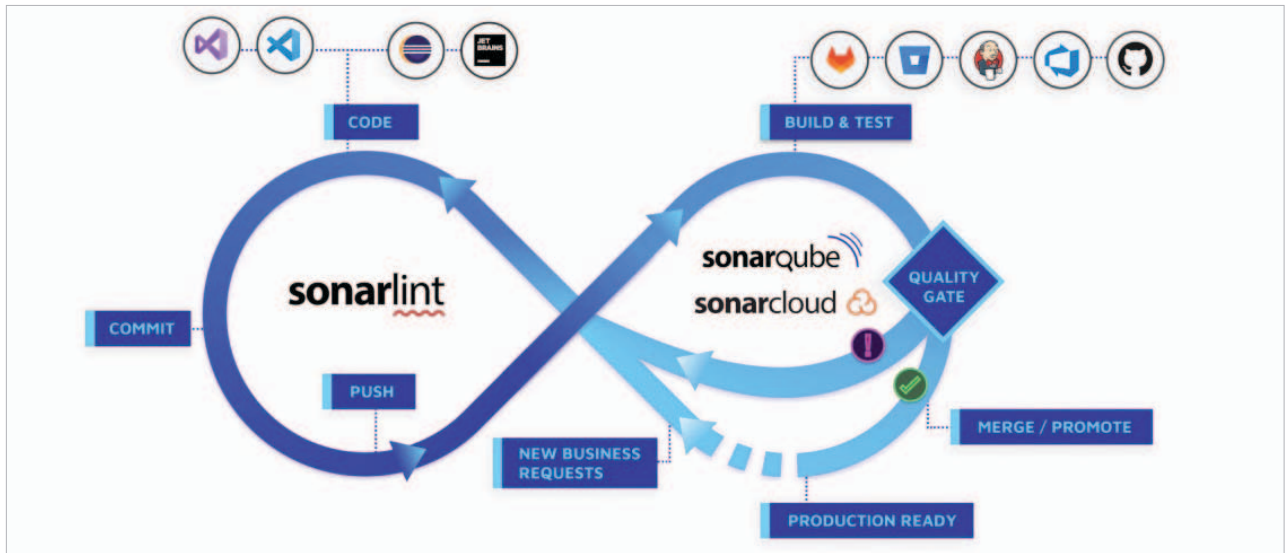


Figure 1: Developing with Sonar (Image: <https://docs.sonarqube.org/latest/>)

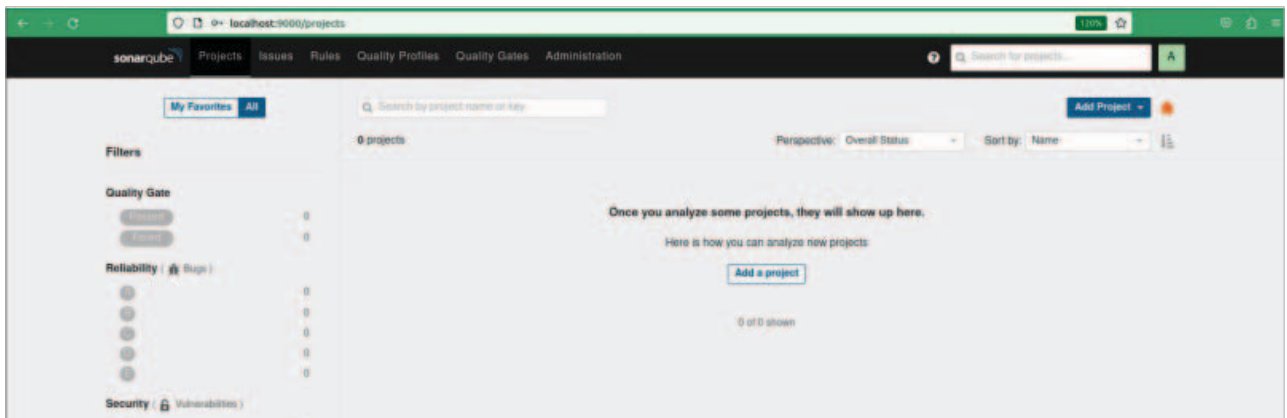


Figure 2: SonarQube dashboard

details of code quality and technical debt. The important thing is that it performs static code analysis. It analyses the source code of an application without running it.

Additionally, SonarQube provides a comprehensive set of reports and metrics for tracking and monitoring code quality over time. These metrics include code complexity, code duplication, test coverage, and so on. SonarQube also offers plugin architecture. Overall, SonarQube is a valuable tool for any development team that wants to improve the code quality and maintainability of its codebase. By automating code inspection by integrating it with CI/CD tools, SonarQube can help teams to deliver better applications faster and with fewer issues.

How to install SonarQube

Here are the basic steps to install SonarQube on a Windows or Linux machine:

1. Download the latest version of SonarQube. Go to the

SonarQube official website <https://www.sonarsource.com/open-source-editions/> and download the latest version of the community edition from <https://www.sonarsource.com/products/sonarqube/downloads/>.

2. SonarQube requires Java 8 or higher to be installed. Download Java from the official Oracle website.

```
osfy@ubuntu:~/Desktop/OSFY/March2023/#tools/
sonarqube-8.9.10.61524/bin/linux-x86-64$ java --version
openjdk 11.0.17 2022-10-18
OpenJDK Runtime Environment (build 11.0.17+8-post-Ubuntu-1ubuntu222.04)
OpenJDK 64-Bit Server VM (build 11.0.17+8-post-Ubuntu-1ubuntu222.04, mixed mode, sharing)
```

3. SonarQube requires a database to store its data, so install one. You can use either a MySQL, PostgreSQL, or Microsoft SQL server. Remember that an embedded